Examiner only

SECTION B

Answer **all** *questions in the spaces provided.*

8. *(a)* Iron(III) oxide, Fe_2O_3, and carbon monoxide react to give iron and carbon dioxide.

Give the equation for this reaction. Explain why it is described as a redox process. [2]

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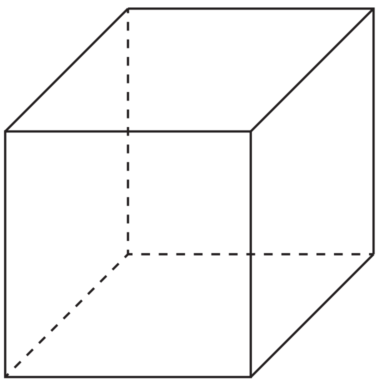
(b) The iron obtained from the blast furnace contains some sulfur as an impurity. This is removed during Basic Oxygen Steelmaking (BOS), by adding magnesium, which removes the sulfur as magnesium sulfide, MgS.

350 tonnes of impure iron used for BOS contain 0.02 % of sulfur. Calculate the mass, in **kilograms**, of magnesium needed to remove all the sulfur. [2]

Mass = ... kg

(c) Magnesium sulfide has the same crystal structure as sodium chloride.

Use the diagram below to show the crystal structure of magnesium sulfide, clearly labelling the formula of each species present. [2]



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(d) Magnesium sulfide reacts with water producing gaseous hydrogen sulfide and magnesium hydroxide.

$$MgS \quad + \quad 2H_2O \longrightarrow Mg(OH)_2 \quad + \quad H_2S$$

(i) State what would be seen during the reaction apart from gas bubbles. [1]

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(ii) A student added a small sample of magnesium sulfide of mass 0.224 g to some water in a fume cupboard.

Calculate the maximum volume, in cm^3, of hydrogen sulfide produced at 25 °C. [3]

Volume = ... cm^3

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(e) When barium sulfide is added to water a similar reaction occurs to that described in part *(d)* with gaseous hydrogen sulfide and barium hydroxide being the products.

State with a reason, how the observation for this reaction would be different from that seen with magnesium sulfide. [2]

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(f) Barium hydroxide can also be produced by adding barium oxide to water.

Give the equation for this reaction and estimate the pH of the product. [2]

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(g) Explain why it is unlikely that stable compounds containing Ba^{3+} ions can exist. [2]

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